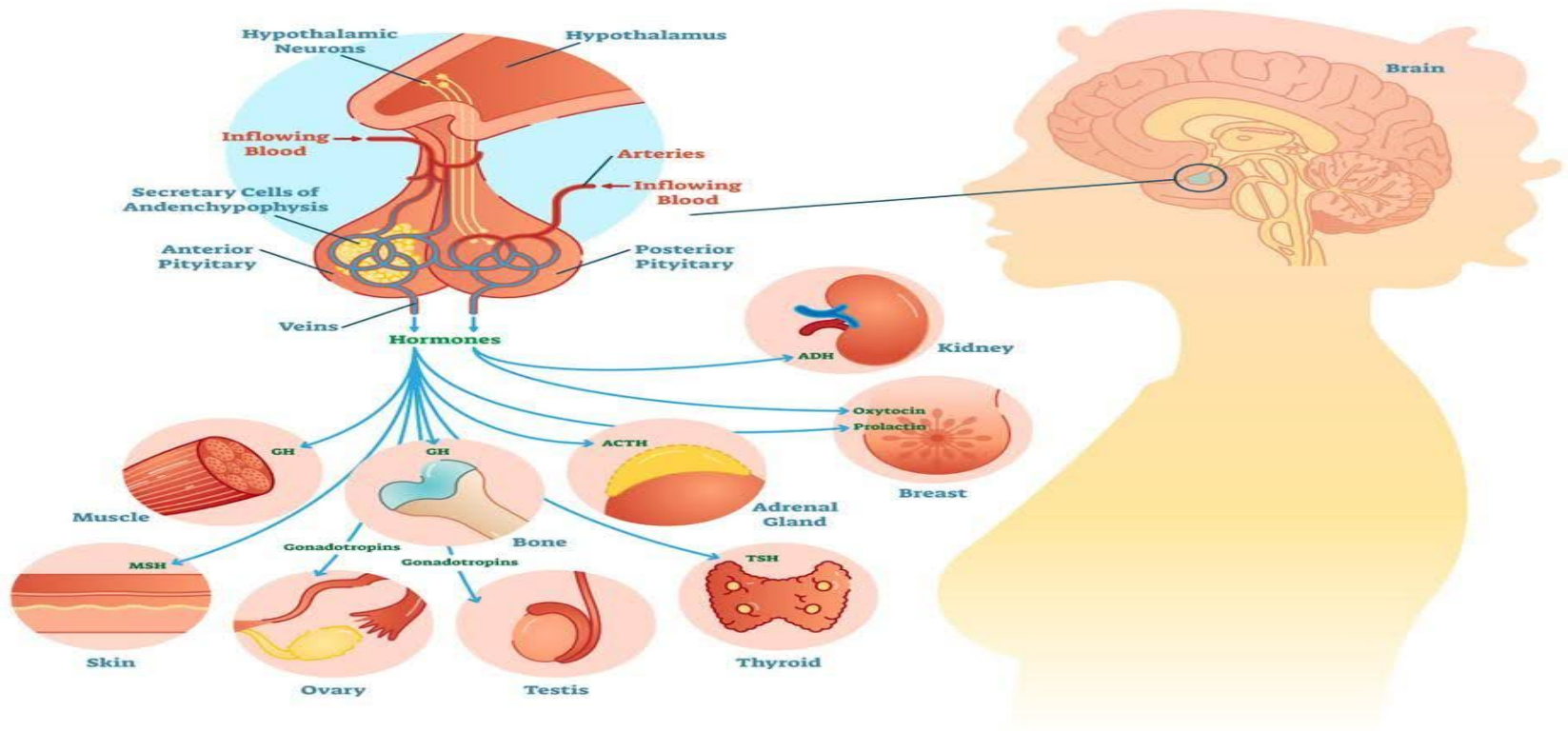


Pituitary and Hypothalamic hormones

PITUITARY GLAND



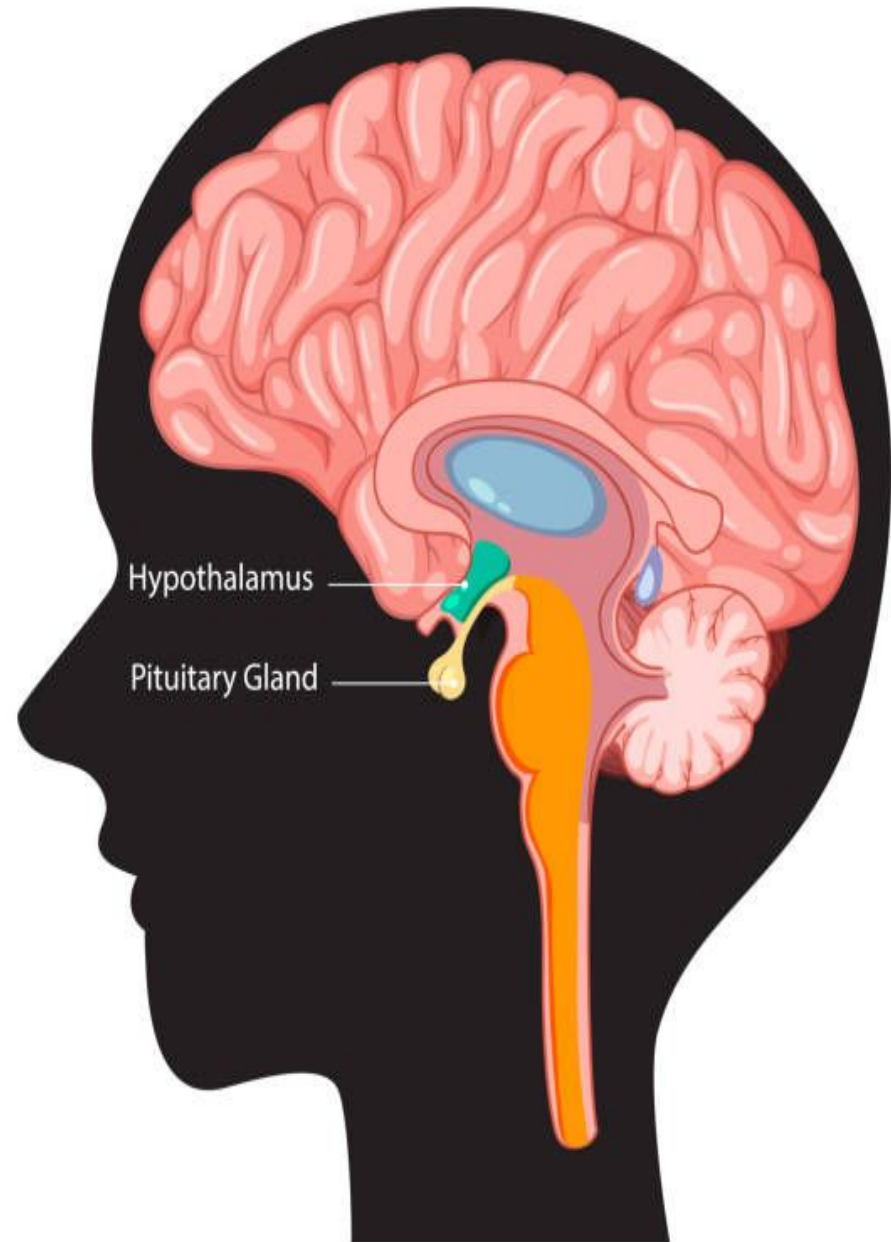
DR ABDUL SAMAD KHAN
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Learning Objectives

- **Describe** pituitary hormones
- **Explain** hypothalamic control
- **Relate** hormones to target glands

What is the hypothalamus?

- The hypothalamus is a structure deep within the brain. It's the main link between endocrine system and nervous system.
- It acts as body's smart control coordinating center.
- Its main function is to keep the body in a stable state called homeostasis
- Many conditions can damage your hypothalamus, which can affect many bodily functions.



Functions of Hypothalamus

Hypothalamus receives chemical messages from nerve cells in the brain and from nerve cells in the body which also respond to signals outside body.

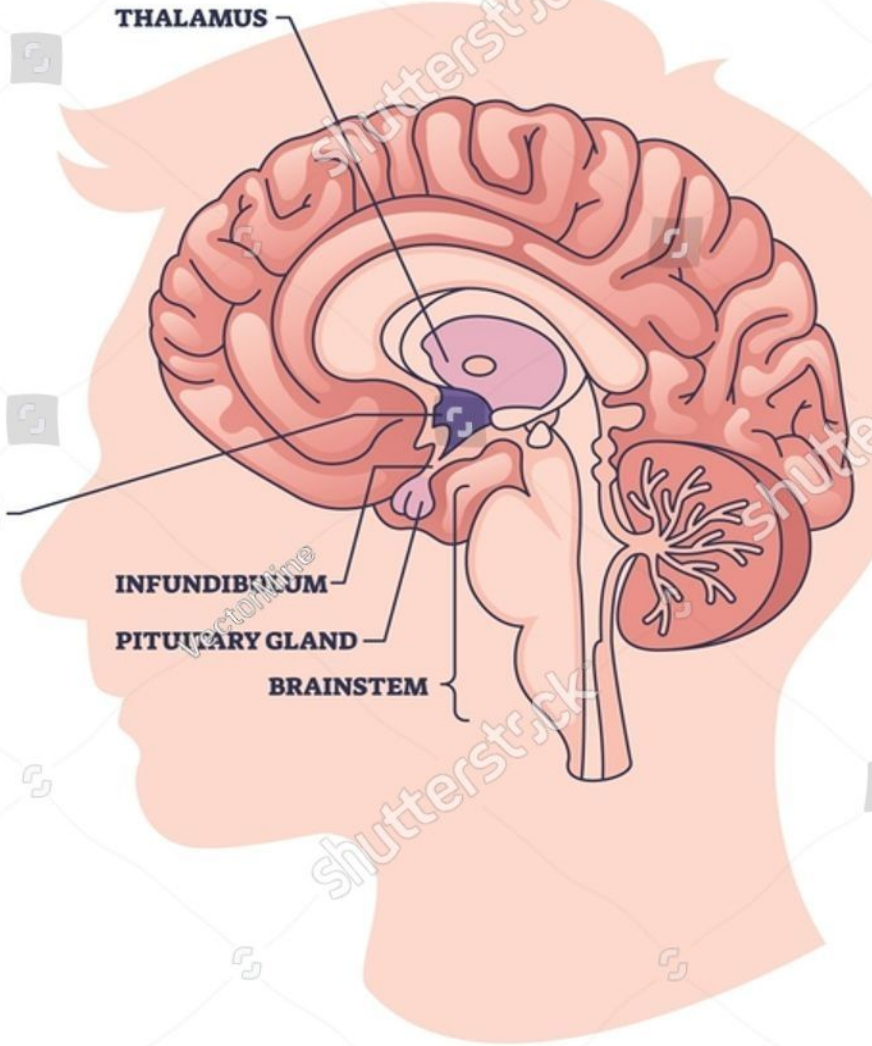
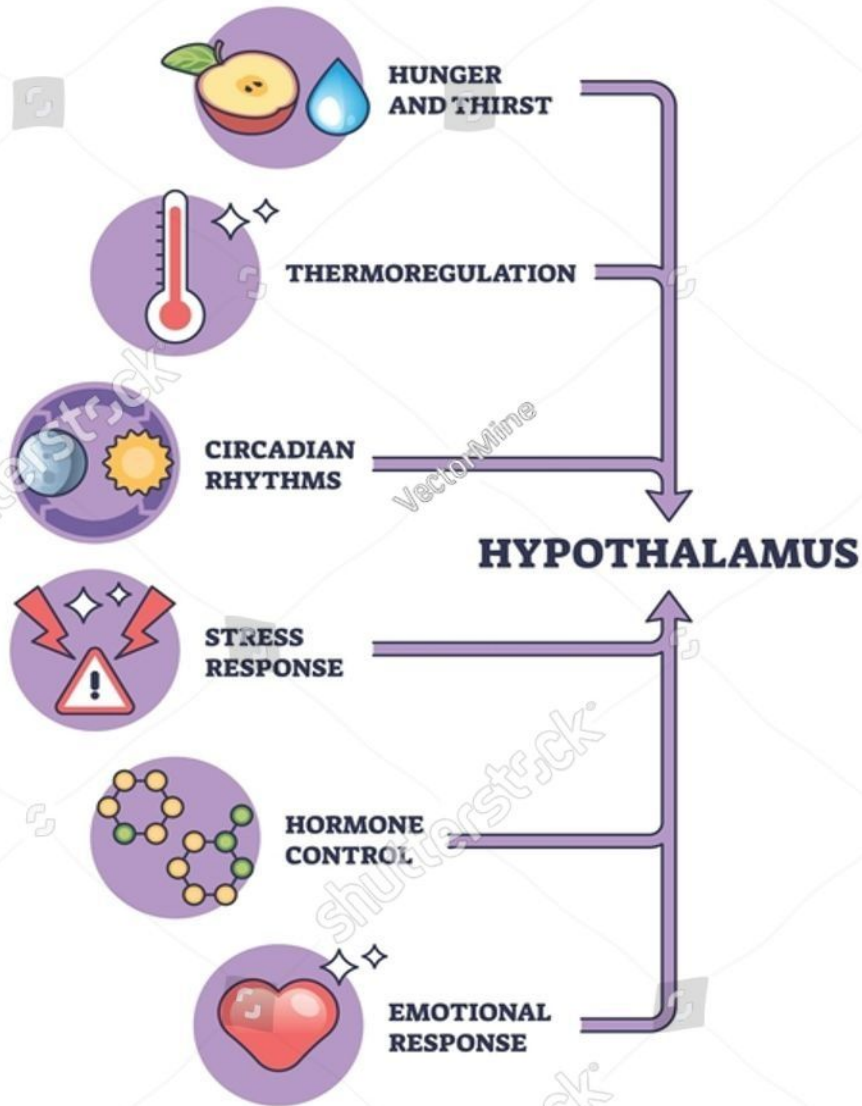
Hypothalamus's main function is to react to these messages to keep your body in a stable state or internal balance.

Hypothalamus is body's "smart control" coordinating center.

Hypothalamus helps to manage following :

- ❑ Body temperature
- ❑ Blood pressure
- ❑ Hunger and thirst
- ❑ Sense of fullness when eating
- ❑ Mood
- ❑ Sex drive
- ❑ Sleep

HYPOTHALAMUS FUNCTIONS



What are the symptoms of hypothalamus dysfunction?

Symptoms of hypothalamus dysfunction correspond to the types of hormone involved and if the hormone level is too low or too high.

Some symptoms of a hypothalamus problem may include:

- ❑ High blood pressure or low blood pressure
- ❑ Water retention or dehydration
- ❑ Weight loss or weight gain with or without changes in appetite
- ❑ Infertility
- ❑ Poor bone health
- ❑ Delayed puberty
- ❑ Muscle loss and weakness
- ❑ Body temperature fluctuations
- ❑ Trouble sleeping insomnia
- ❑ Frequent need to pee



Physical and Mental Stress



Hypothalamic Dysfunction



Orexin Deficiency



Low Adrenal Response



Persistent Neuroinflammation



Glial Cell Reactivity



Conditions that can affect the functions of the hypothalamus

Following conditions can lead to dysfunction of the hypothalamus

Hypothalamic dysfunction can be due to:

- 1- Head injuries, such as a traumatic brain injury
- 2- Brain infection
- 3- Brain tumor
- 4- Brain surgery
- 5- Radiation therapy and chemotherapy
- 6- Birth defects (congenital conditions) involving the brain or hypothalamus
- 7- Inflammatory disease including multiple sclerosis and neurosarcoidosis
- 8- Some genetic disorders, such as growth hormone deficiency

Hormones produced by the hypothalamus

The hypothalamus produces:

Dopamine

It inhibits prolactin secretion

Thyrotropin Releasing hormone

(TRH) which regulates the secretion of thyroid stimulating hormone (TSH)

Corticotropin Releasing hormone (CRH),

It stimulates the secretion of the Adrenocorticotrophic hormone (ACTH)

Growth hormone releasing hormone (GHRH)

It stimulates growth hormone secretion (GH)

Somatostatin

It inhibits the secretion of growth hormone and thyroid stimulating hormone

The gonadotropin releasing hormone (GnRH)

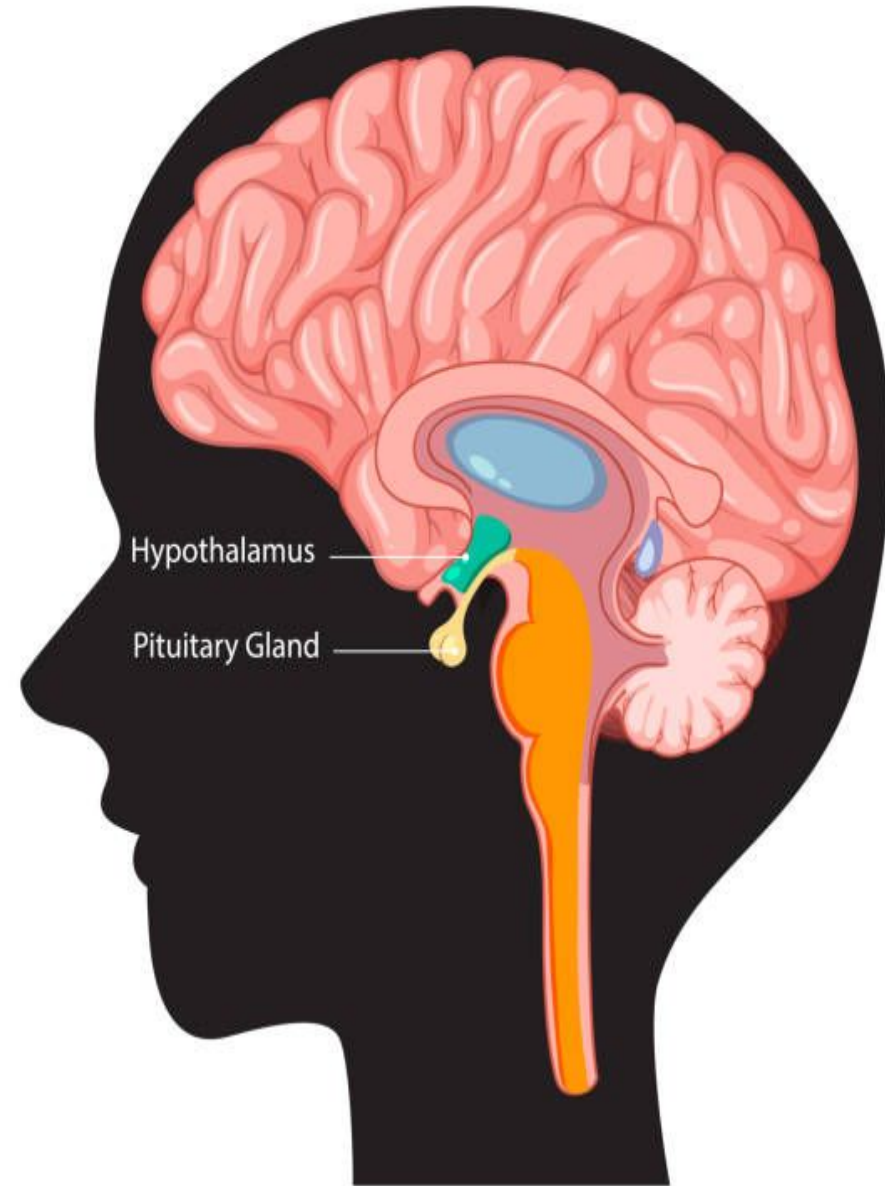
It stimulates the secretion of LH and FSH.

Hypothalamus and the Pituitary gland

The hypothalamus and pituitary gland are located in the brain, and specifically at the base of the skull, behind the eyes.

They form the central unit of coordination of the endocrine system and have specialized cells producing hormones

Each cell produces a different hormone that has a specific rate of secretion.



How does the hypothalamus interact with the pituitary gland?

Hypothalamus is connected to and communicates with the anterior lobe through a network of blood vessels.

It communicates with the posterior lobe by tissue called the **pituitary stalk**.

Pituitary gland sits just below the hypothalamus. It consists of two lobes, called the anterior pituitary and posterior pituitary.

The hypothalamus interacts with the pituitary gland by sending chemical and electrical signals through a connecting stalk called **the infundibulum**

Hypothalamus sends signals in the form of releasing hormones to tell the anterior and posterior pituitary when to release (secrete) its hormones.

Pituitary gland and Hormones

The pituitary is a small, bean-sized gland. It is located at the base of brain below the hypothalamus.

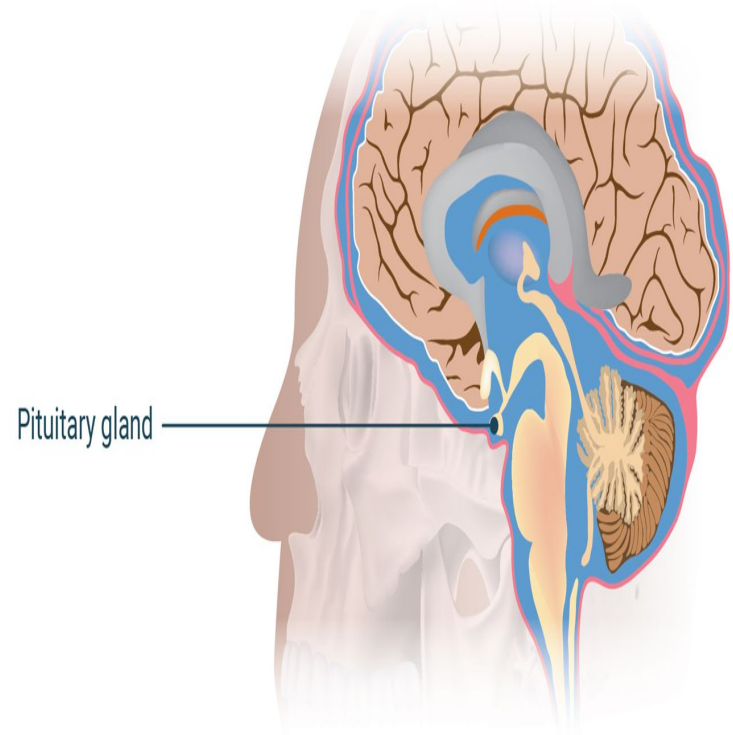
It sits in its own little chamber under your brain known as the **sella turcica**. It is part of endocrine (hormone) system.

Pituitary acts as **body's 'master gland'**.

It secretes hormones that control vital functions and other glands

Pituitary gland is divided into two main sections: the anterior pituitary (front lobe) and the posterior pituitary (back lobe).

It tells glands in body what to do.



Hormones produced by the pituitary gland:

The pituitary gland consists of two parts, the anterior or adenohypophysis, and the posterior or neurohypophysis,

Anterior Pituitary gland secretes:

Prolactin

Responsible for milk production during breastfeeding,

Thyroid stimulating hormone (TSH)

It regulates thyroid hormone production by the thyroid gland

Adrenocorticotrophic hormone (ACTH)

It controls the production of steroid hormones from the adrenal cortex

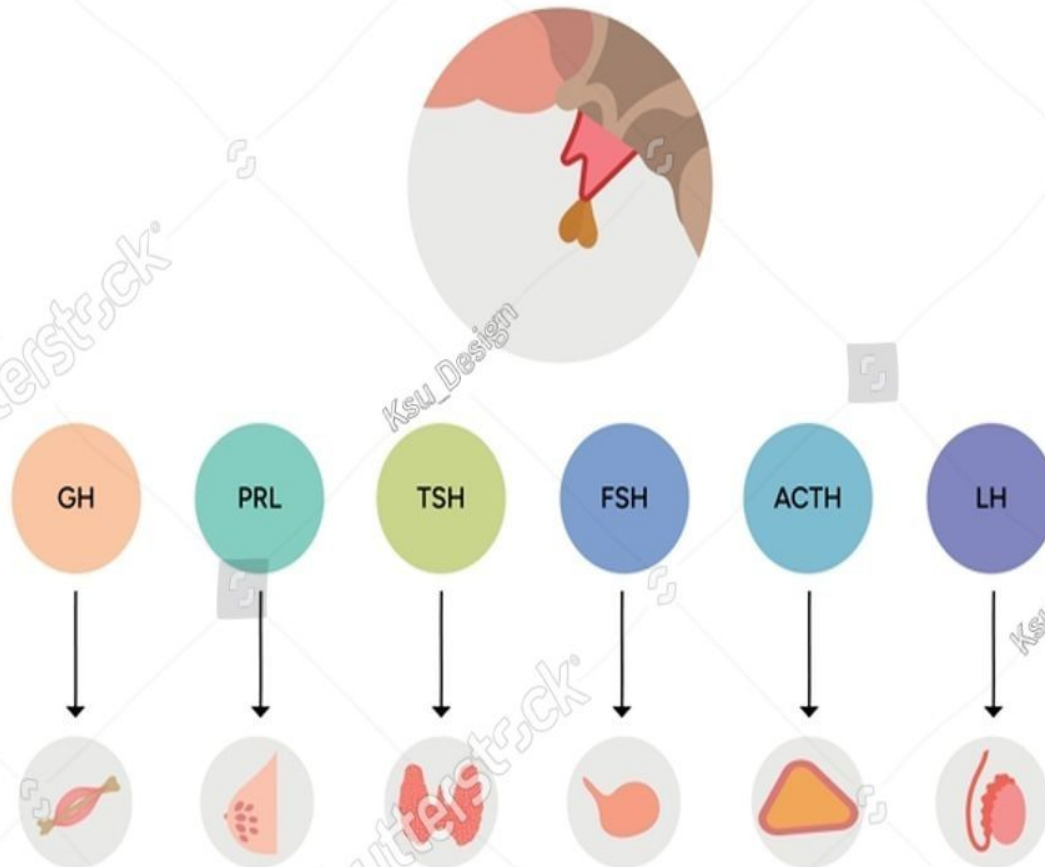
Growth hormone (GH)

Responsible for growth

Gonadotropins (LH, FSH)

It regulate reproductive function

Anterior pituitary gland hormones



Posterior Pituitary gland secretes

The posterior lobe does not make its own hormones. Instead, it stores and releases two hormones made by the hypothalamus

Antidiuretic hormone (ADH)

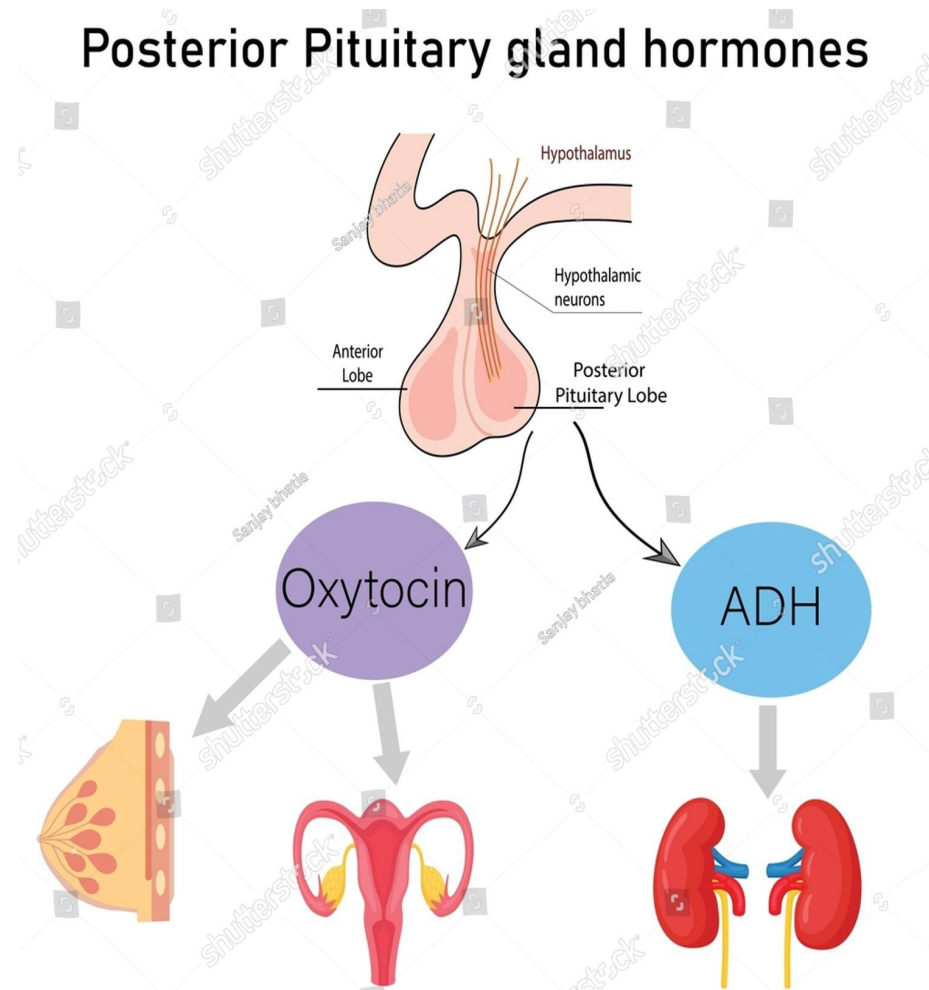
It regulates the homeostasis of water and sodium, condensing urine.

Oxytocin,

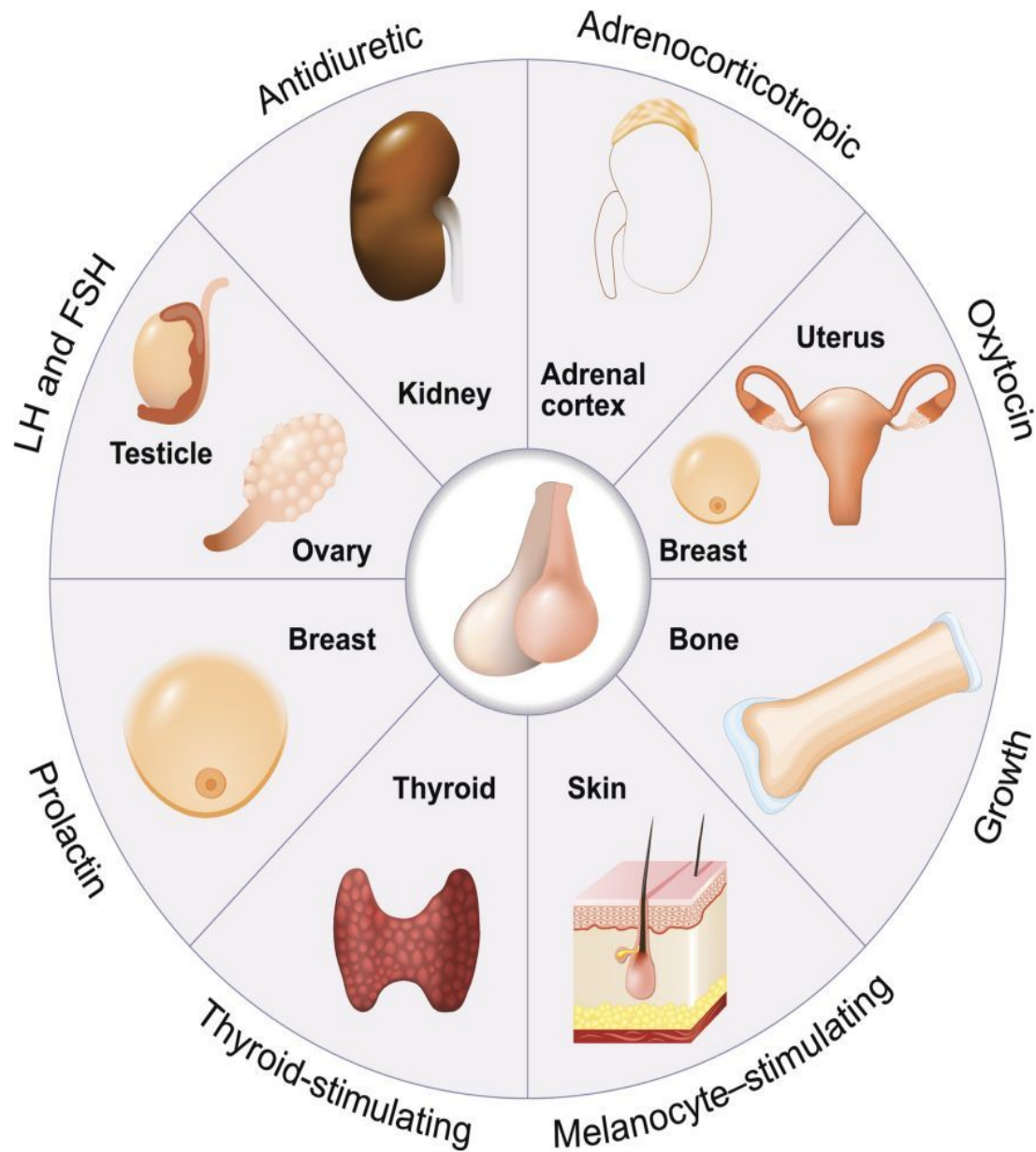
It promotes childbirth and the descent of milk during breastfeeding.

The pituitary hormones are controlled by the hormones of the hypothalamus.

Posterior Pituitary gland hormones



PITUITARY HORMONES



PITUITARY GLAND HORMONES

ANTERIOR PITUITARY

FSH — Follicle-stimulating Hormone
Gametogenesis, estrogen/sperm production

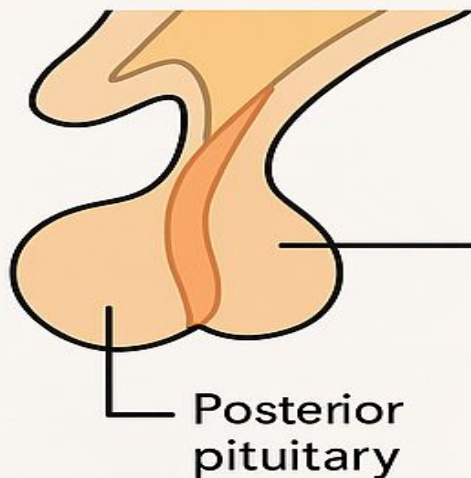
LH — Luteinizing Hormone
Ovulation, testosterone production

ACTH — Adrenocorticotrophic Hormone
Stimulates cortisol secretion

TSH — Thyroid-stimulating Hormone
Stimulates thyroid hormone release

Prolactin Milk production (lactation)

GH — Growth Hormone
Growth, metabolism



POSTERIOR PITUITARY

ADH — Antidiuretic Hormone (Vasopressin)
Water reabsorption, vasoconstriction

Oxytocin
Uterine contractions, milk ejection (letdown reflex)

Control of the Anterior Pituitary

Vascular link: The hypothalamus connects to the front part of the pituitary (the anterior lobe) through a specialized network of blood vessels called the hypophyseal portal system.

Releasing hormones: The hypothalamus releases specific signaling hormones into these blood vessels.

Action: These signals tell the anterior pituitary to produce and release its own hormones into the body, which then control other glands like the thyroid, adrenals, and gonads.

Control of the Posterior Pituitary

Neural link: The hypothalamus connects to the back part of the pituitary (the posterior lobe) directly through nerve fibers.

Direct production: The hypothalamus actually makes two hormones itself—oxytocin and antidiuretic hormone (ADH)—and stores them in the posterior pituitary.

Action: Electrical impulses from the hypothalamus tell the posterior pituitary when to release these stored hormones directly into the bloodstream

FUNCTIONS OF THE PITUITARY GLAND

Pituitary gland makes hormones. It also controls many of the other endocrine glands. The main function of pituitary gland is to produce and release several hormones that help to carry out important bodily functions, including:

- ❑ Growth
- ❑ Metabolism
- ❑ Reproduction
- ❑ Response to stress or trauma
- ❑ Lactation
- ❑ Water and sodium (salt) balance
- ❑ Labor and childbirth
- ❑ Pituitary gland acts like a thermostat.

Pituitary gland monitors body functions in much the same way. Pituitary sends signals to your organs and glands — via its hormones — to tell them what functions are needed and when. The right settings for your body depend on several factors, including your age and sex.

How is the pituitary gland controlled?

Pituitary gland is controlled by hypothalamus. It sends signals to your pituitary gland, telling it when to release hormones.

The hormones made or controlled by your pituitary gland affect vital body functions, including:

- metabolism
- blood pressure
- reproduction

What medical conditions can affect the pituitary gland?

The pituitary gland can sometimes make too much or too little of a hormone. This may lead to a **'pituitary condition'**.

Pituitary conditions are often caused by growths in the gland. These growths are usually benign (not cancerous). They may lead to hormone problems when they take up too much space.

Other causes of pituitary conditions include:

- ☐ Surgery
- ☐ Radiation therapy
- ☐ Autoimmune diseases
- ☐ Infection
- ☐ Brain injury
- ☐ Problems with the hypothalamus

Pituitary Hormones and Target Glands

Anterior Pituitary Hormones

- **Thyroid-Stimulating Hormone (TSH):**
Stimulates the **thyroid gland** to release thyroid hormones that control metabolism.
- **Adrenocorticotrophic Hormone (ACTH):**
Stimulates the **adrenal cortex** (part of the adrenal glands) to release cortisol for stress management.
- **Follicle-Stimulating Hormone (FSH) and Luteinizing Hormone (LH):**
These tropic hormones target the **gonads** (ovaries and testes) to control sex cell production and sex hormone release.
- **Growth Hormone (GH):**
Targets **liver, bones, muscles, and other tissues** to promote overall body growth and cell turnover.
- **Prolactin:**
Targets the **mammary glands** to trigger milk production after childbirth.

Posterior Pituitary Hormones

- **Antidiuretic Hormone (ADH / Vasopressin):**
Targets the **kidneys** to reduce water loss by increasing water reabsorption into the blood.
- **Oxytocin:**
Targets the **uterus and mammary glands** to cause uterine contractions during birth and milk release during nursing.

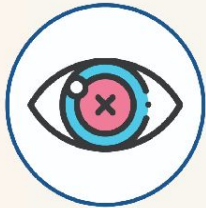
What are Common Symptoms of Pituitary Gland?



Headaches



Growth issue in children



Vision problems



Infertility



Fatigue



Menstrual irregularities



Unexplained weight changes



Causes of Hypopituitarism

Diana Stukalkina,
Kenechukwu Ogbonna,
Owen Hibberd

2025

Congenital

- Genetic diseases
- Idiopathic (when the cause cannot be found)



Acquired

- Pituitary tumours
- Brain tumours (e.g. craniopharyngioma, gliomas)
- Poor blood supply to the pituitary gland
- Infections or inflammatory diseases
- Sarcoidosis
- Amyloidosis
- Autoimmune diseases
- Head injuries,
- Radiation therapy
- Brain surgery
- Frequent blood transfusion



APPLIED PHYSIOLOGY

DISORDERS OF PITUITARY GLAND

HYPERACTIVITY OF ANTERIOR PITUITARY

1. Gigantism

It is the pituitary disorder characterized by excess growth of the body. The subjects look like the giants with average height of about 7 to 8 feet.

2. Acromegaly

It is the disorder characterized by the enlargement, thickening and broadening of bones, particularly in the extremities of the body.

3. Cushing Disease

It is also a rare disease characterized by obesity.

HYPOACTIVITY OF ANTERIOR PITUITARY

1. Dwarfism

Dwarfism is a pituitary disorder in children, characterized by the stunted growth.

2. Acromicria

Acromicria is a rare disease in adults characterized by the atrophy of the extremities of the body.

HYPERACTIVITY OF POSTERIOR PITUITARY

Syndrome of Inappropriate Hypersecretion of Antidiuretic Hormone (SIADH)

SIADH is the disease characterized by loss of sodium through urine due to hypersecretion of ADH.

HYPOACTIVITY OF POSTERIOR PITUITARY

Diabetes insipidus

Diabetes insipidus is a posterior pituitary disorder characterized by excess excretion of water through urine.

Anterior Pituitary Disorders

Hormone	Increased level	Decreased level
GH	Gigantism (child) Acromegaly (adult)	Dwarfism (child) Lethargy, premature aging
ACTH	Cushing's Disease	Addison's Disease
TSH	Goiter, increased BMR, HR, BP Graves disease	Decreased BMR, HR, CO, BP Cretinism (children)
Prolactin	amenorrhea	Too little milk
FSH		Late puberty, infertility
LH	Menstrual cycle disturbance	Amenorrhea, impotence

Posterior Pituitary Disorders

Hormone	Increased	Decreased
Oxytocin	Precipitates childbirth, excess milk	Prolonged childbirth, diminished milk
ADH (vassopressin)	Increased BP, decreased urinary output, edema. SIADH	Diabetes insipidus, dilute urine & increased urine output

Hypothalamus dysfunction plays a role in:

Hypothalamic-pituitary disorders. Because of the close interactions between hypothalamus and pituitary gland, conditions that affect either are called hypothalamic-pituitary disorders. Certain hormone tests can help determine the exact hormone involved to make a more specific diagnosis.

Hypopituitarism. This condition can be caused by damage to the pituitary gland or to hypothalamus, which directly affects the pituitary gland.

Diabetes insipidus. This condition happens when hypothalamus doesn't produce and release enough vasopressin. This causes your kidneys to lose too much water and results in excessive peeing and thirst.

Prader-Willi syndrome. This inherited disorder causes hypothalamus not to recognize the sensation that you're full when you're eating. Without this sensation, you have a constant urge to eat and are at risk for obesity.

Kallmann syndrome. This syndrome has a genetic link to hypothalamic disease, causing such hypothalamic problems in children as delayed or no puberty.

Acromegaly and pituitary gigantism. These are rare disorders of growth due to excessive release of growth hormone from your pituitary gland.

Syndrome of inappropriate antidiuretic hormone. This occurs when the antidiuretic hormone level is too high. Causes include stroke, hemorrhage, infection, trauma, cancer and certain medications.

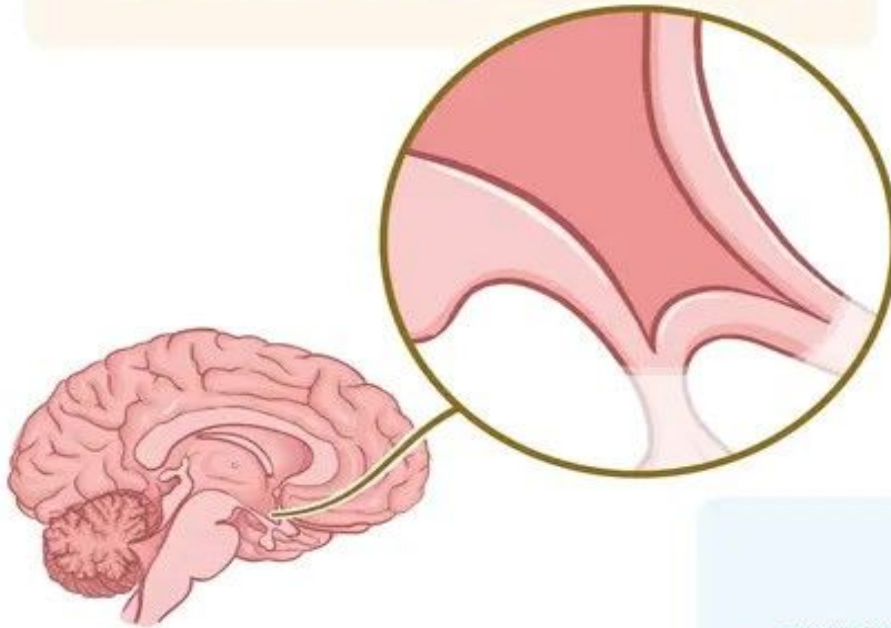
Central hypothyroidism. Central hypothyroidism is a rare disorder that occurs due to both hypothalamic and pituitary disorders. The most common cause is a pituitary tumor such as a **pituitary adenoma**.

Functional hypothalamic amenorrhea. This condition is the absence of a period for more than three months if you're a female who previously had regular periods or more than six months if you have **irregular menstruation**. The most common cause of this condition is hypothalamic dysfunction.

Hyperprolactinemia. A decrease in dopamine levels cause an increase in prolactin levels. Causes include a tumor, damage to nerve cells in the hypothalamus and other causes.

BACKGROUND

- * **IMPORTANT ROLE** in HOMEOSTASIS
- * **PART** of the **DIENCEPHALON** LOCATED between the **TWO CEREBRAL HEMISPHERES**



DISORDERS

- * **GENETIC CONDITIONS**
 - ~ KALLMAN SYNDROME
 - ~ PRADER WILLI SYNDROME
- * **HYPOTHALAMIC OBESITY**
- * **DIABETES INSIPIDUS**
- * **SIADH**



FUNCTION

* **REGULATE BODILY PROCESSES**

- ~ **THIRST & WATER INTAKE**
- ~ **HUNGER & FOOD INTAKE**
- ~ **AUTONOMIC NERVOUS SYSTEM**
- ~ **BIOLOGICAL CIRCADIAN RHYTHMS**
- ~ **BODY TEMPERATURE**
- ~ **BLOOD PRESSURE**
- ~ **BREAST FEEDING**
- ~ **LEARNING & MEMORY**
- ~ **SEXUAL DRIVE**
- ~ **EMOTIONAL EXPRESSION**



* **PRODUCE HORMONES**

HORMONES PRODUCED

- * **THYROTROPIN-RELEASING HORMONE (TRH)**
- * **CORTICOTROPIN-RELEASING HORMONE (CRH)**
- * **GONADOTROPIN-RELEASING HORMONE (GnRH)**
- * **GROWTH HORMONE-RELEASING HORMONE (GHRH)**
- * **SOMATOSTATIN**
- * **DOPAMINE**
- * **VASOPRESSIN/ANTIDIURETIC HORMONE (ADH)**
- * **OXYTOCIN**



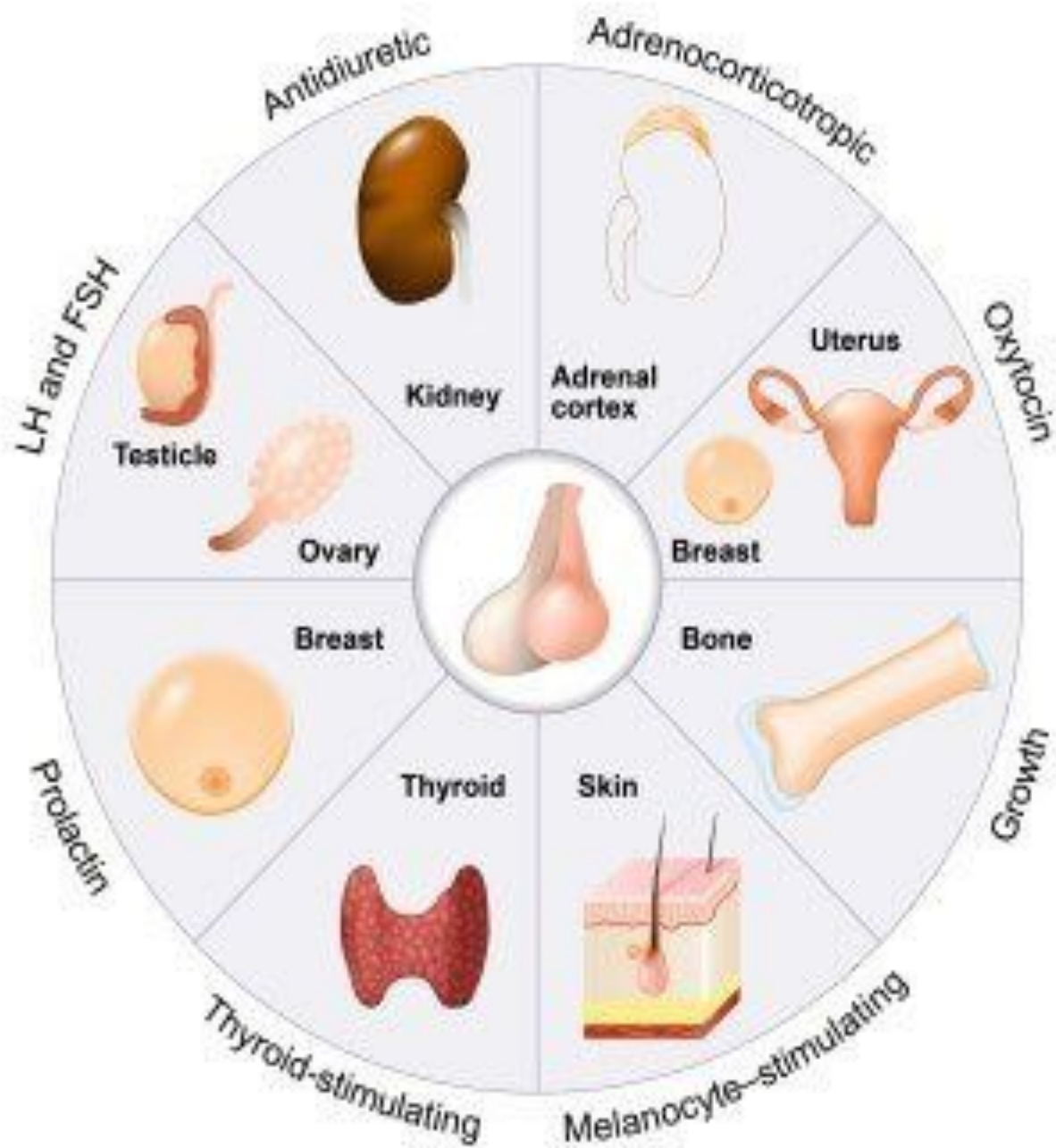
Hypothyroidism

- Hair loss
- Inability to think clearly
- Goiter (enlarged thyroid)
- Reduced heart rate
- Strong fatigue
- Sensitivity to cold
- Dry skin
- Weight gain
- Puffiness
- Memory problems
- Constipation
- Irregular menstrual periods
- Severe PMS
- Depression, mood swings
- Joint, muscle pain
- High cholesterol

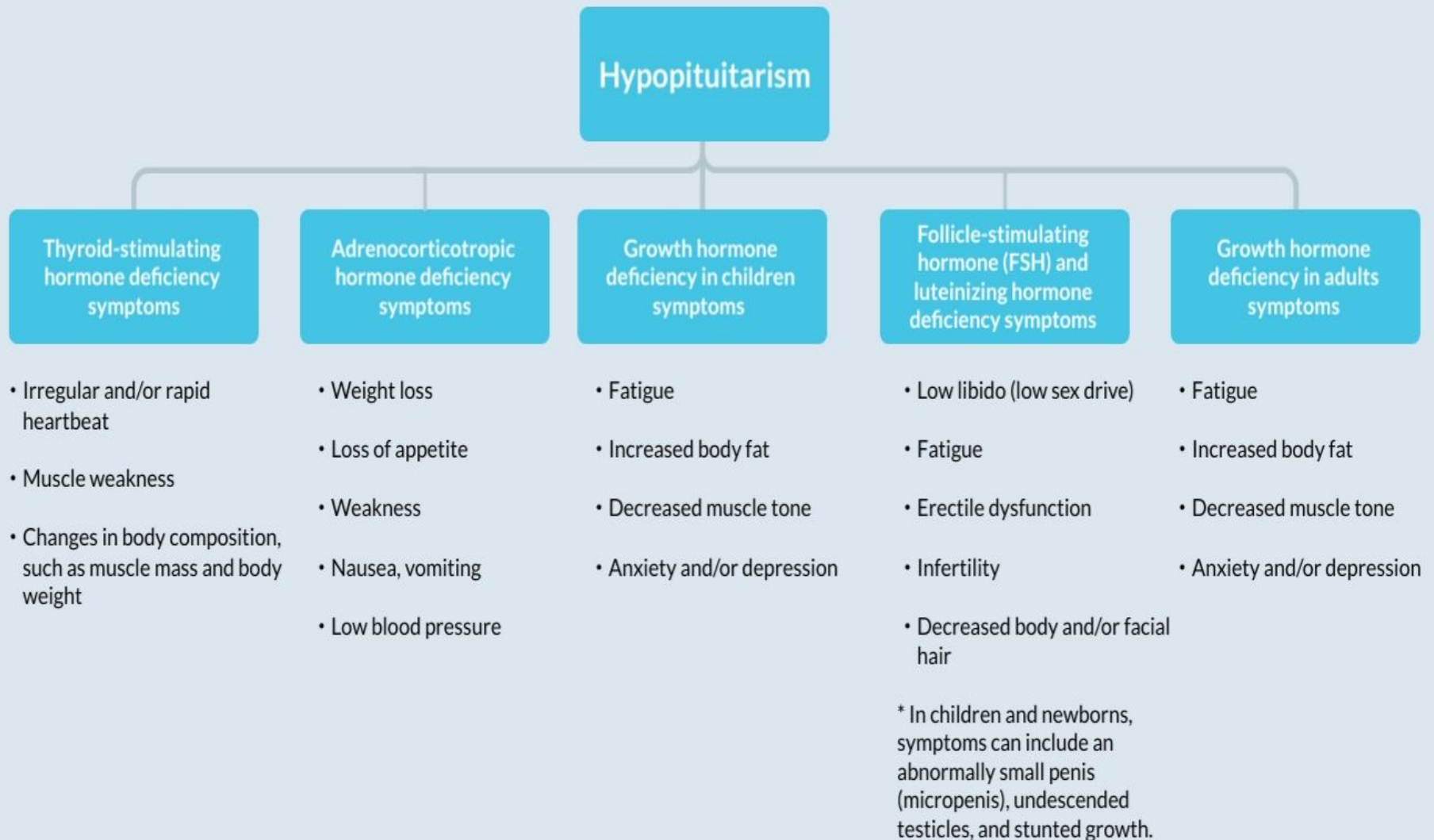


Hyperthyroidism

- Hair loss
- Bulging eyes
- Goiter (enlarged thyroid)
- Heart palpitations
- Tremors
- Heat intolerance
- Sleep disturbances
- Weight loss
- Shortness of breath
- Diarrhoea
- Increased appetite
- Irregular menstrual periods
- Muscle weakness
- Sweating
- Anxiety, nervousness
- Depression, mood swings



Symptoms of hypopituitarism



Hypopituitarism

SIGNS/SYMPTOMS BY HORMONE

GROWTH HORMONE

Children

- Slow growth, delayed puberty



Adults

- Reduced muscle & bone.
- Increased fat mass
- Risk CV disease, insulin resistance.



LH/FSH

- Amenorrhea, ED, infertility.
- Kallmann Syndrome: genetic absence of GnRH receptors.



THYROID STIMULATING HORM.

- Hypothyroidism
- Hypometabolism; fatigue, depression, Impaired CV, Resp.



ACTH

- Central AI: Low Cortisol & Adrenal Androgens



PRL

- Deficient lactation

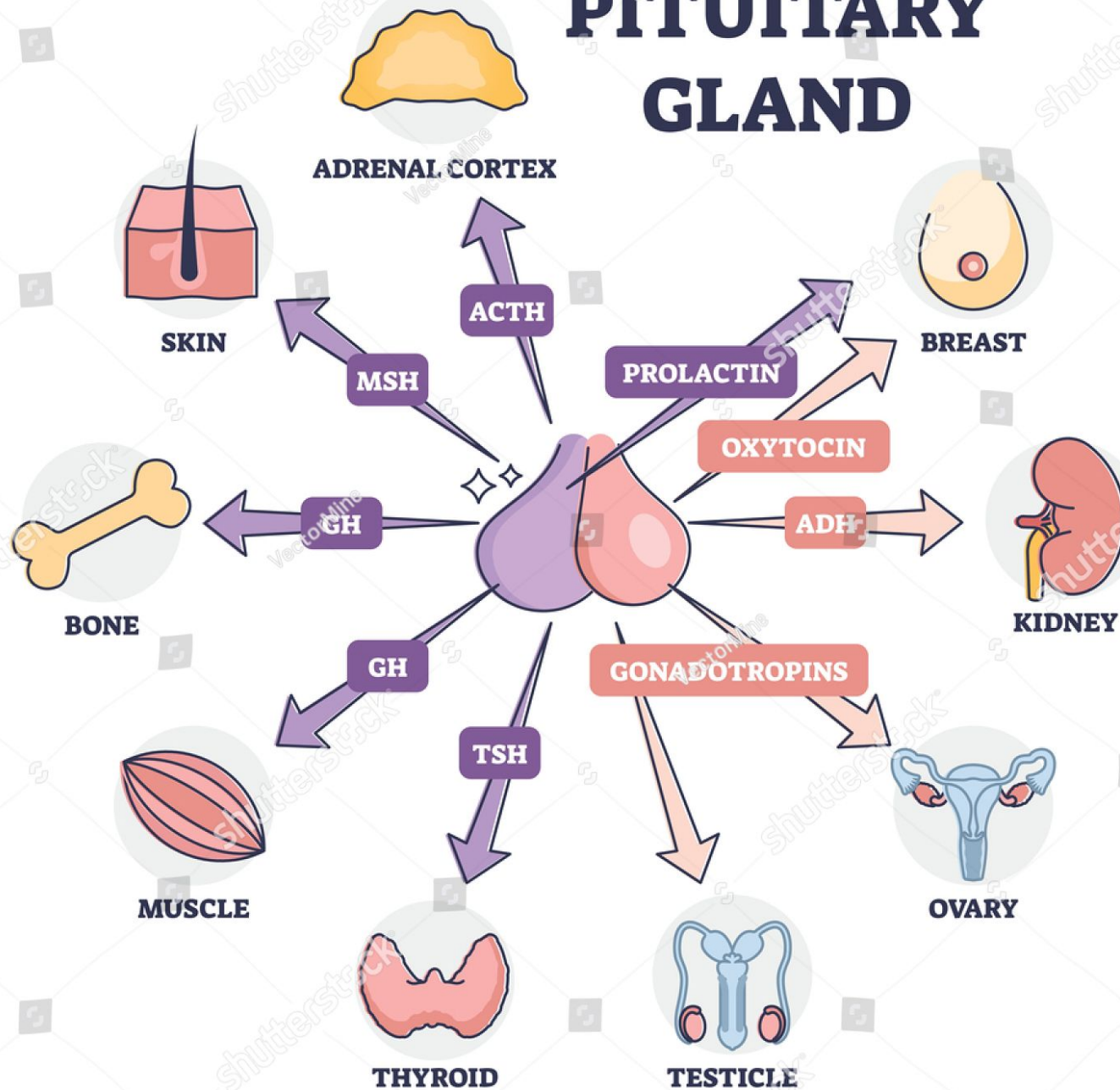


ADH (POST. PIT)

- Diabetes insipidus: Dehydration.



PITUITARY GLAND



Pituitary gland

